

Disproof of conjecture 00000008433: no threshold in t alone makes the divisibility conditions sufficient for t -designs

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1 The conjecture and the reading

Statement (English, verbatim). Definition: The Wilson asymptotics of t -designs: their existence for large v . Conjecture: Once v is at least $\exp(\exp(t))$, existence is decided solely by the divisibility conditions; and the bound can be improved to v at least t^{ct} with c an absolute constant. (improvement of the Wilson existence bound)

The Chinese version says the same: after $v \geq \exp(\exp(t))$ existence is determined only by the divisibility conditions, and the bound can be improved to $v \geq t^{ct}$ with c an absolute constant.

Reading. Both thresholds, $\exp(\exp(t))$ and t^{ct} , are functions of t alone, and c is called *absolute*. We therefore read each clause as: for all parameters t, v, k, λ (non-degenerate, see Definition 3) with v at least the threshold, a t -(v, k, λ) design exists if and only if the divisibility conditions hold. The second clause is read as “there is a real constant c such that ...”. We refute both clauses, for every real c , and more generally every threshold $T(t)$ that depends on t only.

Not refuted. The reading in which the threshold may also depend on k and λ (a bound $v \geq v_0(t, k, \lambda)$) is not refuted here. That reading replaces the stated $\exp(\exp(t))$ and t^{ct} by different functions, and our family has $k \rightarrow \infty$.

2 Definitions (as in the Lean file)

Following the standard definition (quoted from Wikipedia, *Block design*, retrieved 2026-10-05): “Given any positive integer t , a t -design B is a class of k -element subsets of X , called blocks, such that every point x in X appears in exactly r blocks, and every t -element subset T appears in exactly λ blocks.” The same page says “Block designs may or may not have repeated blocks. Designs without repeated blocks are called simple”.

Definition 1 (`IsDesign`, `HasDesign`, `HasSimpleDesign`). Let α be a finite point set and ι a finite index set. A family $\text{blk} : \iota \rightarrow \text{Finset } \alpha$ is a t -(v, k, λ) design ($v = |\alpha|$) if $|\text{blk}(i)| = k$ for every i , and every t -subset $T \subseteq \alpha$ satisfies $\#\{i : T \subseteq \text{blk}(i)\} = \lambda$. Blocks are counted by index, so repeated blocks are allowed. `HasDesign t v k lam` means that such a family exists on $\text{Fin } v$ for some number b of blocks. `HasSimpleDesign` asks for a set D of k -subsets with each t -subset in exactly λ members of D . The condition “every point is in exactly r blocks” is not imposed. Omitting it only enlarges the class of designs, so non-existence proved without it also covers definitions that include it.

Definition 2 (DivCond). The divisibility conditions: $\binom{k-i}{t-i}$ divides $\lambda \binom{v-i}{t-i}$ for $i = 0, 1, \dots, t$. (The case $i = t$ is trivial, so this is equivalent to the version with $i < t$.)

Definition 3 (Admissible). $2 \leq t < k$, $k + t < v$, and $1 \leq \lambda \leq \binom{v-t}{k-t}$. These conditions exclude trivial and degenerate parameter sets.

Clause1: for all t, v, k, λ with **Admissible** and $\exp(\exp t) \leq v$, **HasDesign** $\iff$ **DivCond**.
Clause2: there is $c \in \mathbb{R}$ such that for all admissible t, v, k, λ with $t^{ct} \leq v$ (real power), **HasDesign** $\iff$ **DivCond**.

3 The counterexample family

For $m \geq 3$ put $t = 2$, $\lambda = 1$, $v = (m-1)^2(m+1)$ and $k = m(m-1)$ (Lean: **vF**, **kF**). Then $v-1 = m(m^2-m-1) = m(k-1)$, so $r = (v-1)/(k-1) = m$, and $v(v-1)/(k(k-1)) = (m-1)(m+1) = m^2-1$. For example, $m = 13$ gives 2-(2016, 156, 1), and $2016 > e^{e^2} \approx 1618.18$ (this numerical remark is not used in the proof).

Lemma 4 (**family_admissible**, **family_divCond**). For $m \geq 3$ the parameters are admissible and satisfy all divisibility conditions: $i = 0$: $\binom{k}{2}(m^2-1) = \binom{v}{2}$; $i = 1$: $(k-1)m = v-1$; $i = 2$: trivial.

Proof. Write $m = p+3$. Then $v = (p+2)^2(p+4)$, $k = (p+3)(p+2)$, $k-1 = p^2+5p+5$ and $v-1 = (p+3)(p^2+5p+5)$. The identities follow by ring arithmetic, using $2\binom{n}{2} = n(n-1)$.
 Admissibility: $k \geq 6 > 2$, $k+2 < v$, and $\binom{v-2}{k-2} \geq 1$ because $k \leq v$. $\square$

Lemma 5 (**fisher_bound**). Let $\text{blk} : \iota \rightarrow \text{Finset } \alpha$ be a 2-($v, k, 1$) design with $2 \leq k < v = |\alpha|$. Then $k(k-1) \leq v-1$.

Proof. This is the $\lambda = 1$ case of Fisher's inequality $b \geq v$ (Wikipedia, *Block design*: "Fisher's inequality, named after the statistician Ronald Fisher, is that $b \geq v$ in any 2-design"). We prove it directly by two double counts. Since $v \geq 2$, some pair lies in a block $B_0 = \text{blk}(i_0)$, and since $|B_0| = k < v$ there is a point $x \notin B_0$. Let $S = \{i : x \in \text{blk}(i)\}$. (a) $k \leq |S|$: each $y \in B_0$ lies in some block of S , namely the block through the pair $\{x, y\}$. A block $i \in S$ contains at most one point of B_0 . Otherwise two points $y \neq z$ of B_0 would lie in both $\text{blk}(i)$ and B_0 , and $i \neq i_0$ because $x \notin B_0$. This contradicts $\lambda = 1$. Double counting the incidences (y, i) with $y \in B_0$, $i \in S$ and $y \in \text{blk}(i)$ gives $k \cdot 1 \leq |S| \cdot 1$. (b) $|S|(k-1) \leq v-1$: each $i \in S$ contains $k-1$ points other than x . Each $y \neq x$ lies in at most one block of S , since such blocks contain $\{x, y\}$. Double counting gives the bound. Hence $k(k-1) \leq |S|(k-1) \leq v-1$. Both counts use Mathlib's **Finset.card_mul_le_card_mul**. $\square$

Theorem 6 (**family_no_design**, **family_no_simple_design**). For $m \geq 3$ there is no 2-(($m-1$)²($m+1$), $m(m-1)$, 1) design, simple or with repeated blocks.

Proof. By the lemma, $k(k-1) \leq v-1 = m(k-1)$, and $k-1 > 0$, so $k \leq m$. But $k = m(m-1) \geq 2m > m$. A simple design D is the indexed family $D \rightarrow \text{Finset}(\text{Fin } v)$, $B \mapsto B$, with the same incidence counts, so it is excluded as well. $\square$

Theorem 7 (**exists_counterexample**, **threshold_fails**). For every $T : \mathbb{N} \rightarrow \mathbb{R}$ there are admissible t, v, k, λ with $T(t) < v$, the divisibility conditions satisfied, and no design (indexed or simple). Hence " $\forall t, v, k, \lambda$ admissible with $T(t) \leq v$: **DivCond** $\Rightarrow$ **HasDesign**" is false.

Proof. Choose $N \in \mathbb{N}$ with $N > T(2)$ and take $m = N+3$. Then $v = (N+2)^2(N+4) \geq N$. $\square$

Theorem 8 (`conjecture8433_false`). $\neg \text{Clause1} \wedge \neg \text{Clause2}$. *The first part is `threshold_fails` with $T(t) = \exp(\exp t)$. The second applies `threshold_fails` with $T(t) = t^{ct}$ to each real c .*

`witness_13` records the instance $m = 13$: `vF 13 = 2016, kF 13 = 156, DivCond 2 2016 156 1` and `¬HasDesign 2 2016 156 1`.

4 Lean formalization and axiom audit

The file `lean/Conjecture8433/Basic.lean` (254 lines, only `import Mathlib, Lean v4.33.1, Mathlib v4.33.1`) is built with `lake build`. The theorems named above are in the namespace `Conjecture8433`. The output of `#print axioms` for `conjecture8433_false`, `exists_counterexample`, `threshold_fails`, `witness_13` and `fisher_bound` is `[propext, Classical.choice, Quot.sound]` in each case. The file uses no `sorry`, no `native_decide` and adds no axioms.

Source

Wikipedia, *Block design*, https://en.wikipedia.org/wiki/Block_design (retrieved 2026-10-05), quoted above.